Faculty of Engineering

Module Title: Engineering Chemistry

Sheet 4

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1. Choose the correct answer:

- **From the below options, choose the correct example for gaseous solutions.**
  - Oxygen dissolved in water
  - Natural gas
  - Carbon dioxide dissolved in water

- **Which among the following is an example of a solid solution?**
  - Copper dissolved in gold
  - Ethanol dissolved in water
  - Glucose dissolved in water
  - Sodium chloride dissolved in water

- **The solubility of a substance in a solvent depends on**
  - Temperature
  - Pressure
  - Nature of solute and solvent
  - All of the above

- **How does the solubility of gases vary with pressure?**
  - Increases with pressure
  - Decreases with pressure
  - First increases and then decreases
  - No effect

- **How does the solubility of gases in a liquid vary with temperature elevation?**
  - Increases with temperature
  - Decreases with temperature
  - First increases and then decreases
  - No effect

- **Which law explained the solubility of gases in a liquid?**
  - Charles law
  - Henry's law
  - Raoult's law
  - Boyle's law

- **What does Henry's constant depend upon?**
  - Nature of gas
  - Nature of solvent
  - Temperature
  - All of the above

2. How would you prepare a 0.150 M glucose in water solution?

3. Vinyl chloride (H_2C= CHCl), has a Henry's-law constant of 0.091 mol/ (L. atm) at 25 °C. What is the solubility of vinyl chloride in water in mol/L at 25 °C and a partial pressure of 0.75 atm?

4. The nonstick polymer Teflon is made from tetrafluoro ethylene (C_2F_4). If C_2F_4 is a gas that dissolves in water at 298 K to the extent of $1.01 * 10^{-3}$ M with a partial pressure of 0.63 atm, what is Henry's constant at 298 K?